The spin-1 Haldane gap proof in tensor-network diagrams

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We give a short diagrammatic rederivation of OpenAI's proof of the Haldane gap for the periodic spin-1 Heisenberg chain [1].

With $J = k_B = 1$, the periodic spin-1 antiferromagnetic Heisenberg Hamiltonian is

$$H_L = \sum_{j=1}^{L} \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}, \qquad \boldsymbol{S}_{L+1} = \boldsymbol{S}_1. \tag{1}$$

All lengths are even and at least four. Write the energies, including multiplicities, as $E_0(L) \le E_1(L) \le \cdots$, and let $\Delta_L = E_1(L) - E_0(L)$. We seek a positive lower bound on Δ_L independent of L.

From purity to a gap.— For inverse temperature $\beta > 0$, define

$$\rho_L(\beta) = \frac{e^{-\beta H_L}}{Z_L(\beta)}, \qquad Z_L(\beta) = \operatorname{Tr} e^{-\beta H_L}. \tag{2}$$

Its eigenvalues $w_m = e^{-\beta E_m(L)}/Z_L(\beta)$ satisfy $w_0 \ge w_1 \ge \cdots$. The Gibbs-state purity

$$P_\tau(L,\beta) = \operatorname{Tr} \rho_L(\beta)^2 = \frac{Z_L(2\beta)}{Z_L(\beta)^2} \tag{3}$$

satisfies $P_\tau = \sum_m w_m^2 \le w_0 \sum_m w_m = w_0$. Thus $1 - P_\tau$ bounds the weight outside a ground-state eigenvector, and

$$e^{-\beta\Delta_L} = \frac{w_1}{w_0} \le \frac{1-w_0}{w_0} \le \frac{1-P_\tau(L,\beta)}{P_\tau(L,\beta)}. \tag{4}$$

Our task is therefore to bound $1 - P_\tau$ exponentially in β, with a rate independent of L. We will obtain this bound by doubling L and β together, then filling in the intermediate lengths.

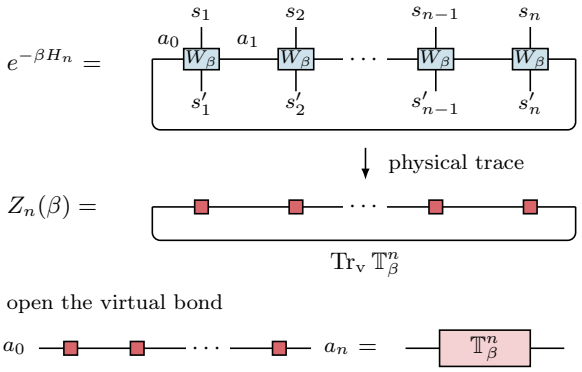
Comparing lengths with an MPO.— To control this purity as L grows, we write Z_L as the trace of a power of one spatial transfer operator. In the Cartesian basis $s, s' \in \{x, y, z\}$, the matrices $(S^\mu)_{ss'} = -i\epsilon_{\mu ss'}$ give real matrices $G_\mu = iS^\mu$, $\|G_\mu\| = 1$, and $-H_L = \sum_{j,\mu} G_{\mu,j} G_{\mu,j+1}$. A bond event (τ, μ) in the exponential expansion inserts G_μ at both endpoints. Let a, b record the events on the two bonds adjacent to a site, with $0 < \tau < \beta$. The local MPO tensor is

$$W_\beta^{ss'}(a,b) = \left\langle s \left| \mathcal{T}_\tau \prod_{(\tau,\mu)\in a\cup b} G_\mu \right| s' \right\rangle, \tag{5}$$

where $\mathcal{T}_\tau$ places later times to the left. Summing event labels and integrating ordered times defines a measure μ_β of mass $e^{3\beta}$, with unit mass for the empty history. Contracting the W_β tensors around the ring recovers the absolutely convergent exponential series. This is an exact MPO on the virtual space $L^2(\mu_\beta)$, with no bond truncation. We use standard MPO diagram conventions [2, 3]; Ref. [3] gives Trotter-based constructions.

Tracing the physical legs gives the real symmetric kernel $\mathbb{T}_\beta(a,b) = \sum_s W_\beta^{ss}(a,b)$. Since $|\mathbb{T}_\beta(a,b)| \le 3$ and μ_β is finite, $\mathbb{T}_\beta$ is self-adjoint and Hilbert–Schmidt. Its eigenvalues $\lambda_i(\beta)$ give, for even L,

$$Z_L(\beta) = \operatorname{Tr}_{\mathrm{v}} \mathbb{T}_\beta^L = \sum_i |\lambda_i(\beta)|^L. \tag{6}$$



The horizontal virtual bond closes with $a_n = a_0$; the vertical physical legs s_j, s'_j remain open. Tracing them gives the red ring. Cutting its closing bond leaves $\mathbb{T}_\beta^n$; closing it takes the virtual trace $\operatorname{Tr}_{\mathrm{v}}$.

This open chain defines a second normalized operator and purity:

$$\begin{aligned} \sigma_n(\beta) &= \frac{\mathbb{T}_\beta^n}{Z_n(\beta)}, \\ P_{\mathrm{s}}(n,\beta) &= \operatorname{Tr}_{\mathrm{v}} \sigma_n(\beta)^2 = \frac{Z_{2n}(\beta)}{Z_n(\beta)^2}. \end{aligned} \tag{7}$$

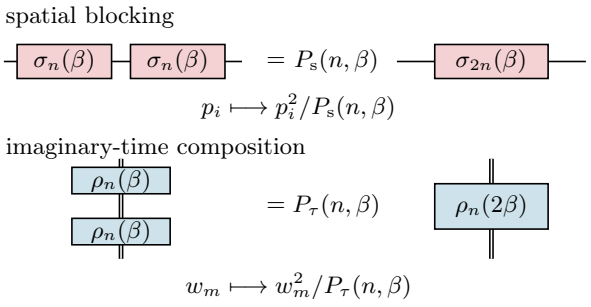
Because n is even, σ_n is positive with unit trace. The transfer purity P_{s} compares lengths, while the Gibbs purity P_τ compares inverse temperatures.

The doubling estimate.— We will propagate the finite-size input using the following implication, valid for $0 \le \varepsilon \le 1/100$:

$$\begin{aligned} & P_{\mathrm{s}}(n,\beta),\ P_\tau(2n,\beta) \ge 1-\varepsilon \\ \Longrightarrow\quad & P_{\mathrm{s}}(2n,2\beta),\ P_\tau(4n,2\beta) \ge 1-5\varepsilon^2. \end{aligned} \tag{8}$$

This doubles both scales and replaces ε by $5\varepsilon^2$. To prove it, we first bound the effect of squaring each operator, then relate the two purities by joining copies of the MPO.

Proof of the doubling estimate.— Let p_i and w_m be the eigenvalues of σ_n and ρ_n. Squaring σ_n doubles n; squaring ρ_n doubles β:

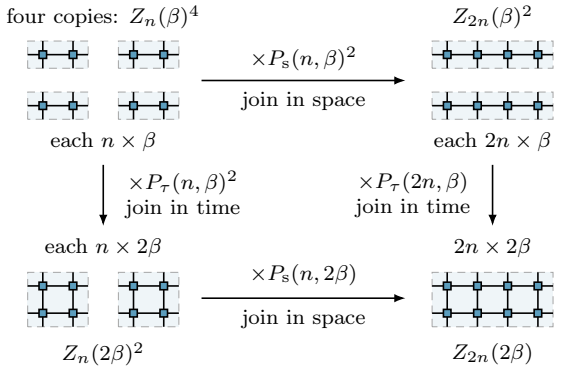


Double legs denote the n-site physical space. For either normalized spectrum, write $P = \sum_i p_i^2$. After $p_i \mapsto p_i^2/P$, its new purity P' satisfies

$$1 - P' = \frac{2\sum_{i<k} p_i^2 p_k^2}{P^2} \le \frac{(1-P)^2}{2P^2} =: f(1-P). \quad (9)$$

Indeed, $1 - P = 2\sum_{i<k} p_i p_k$, and dropping cross terms from the square of this sum gives the inequality. The function f is increasing on $[0,1)$ and behaves as $x^2/2$ for small x.

The squaring identities change one scale at a time. To double both, we need an identity relating P_s and P_τ. Rejoin the trace loops of four copies of $Z_n(\beta)$ in either order:



Each rectangle is periodic in both directions; its width and height denote length and inverse temperature. Tensors are schematic, and arrows give ratios of new to old contraction values. Both orders end at $Z_{2n}(2\beta)$, so equating their ratios gives

$$P_s(n,2\beta) = \frac{P_s(n,\beta)^2 P_\tau(2n,\beta)}{P_\tau(n,\beta)^2}, \quad (10)$$

$$P_\tau(4n,\beta) = \frac{P_\tau(2n,\beta)^2 P_s(2n,2\beta)}{P_s(2n,\beta)^2}. \quad (11)$$

The second line is the same identity at length $2n$, solved for $P_\tau(4n,\beta)$. These identities explain the pair of lengths $n, 2n$ in Eq. (8).

Now assume $P_s(n,\beta), P_\tau(2n,\beta) \ge 1-\varepsilon$. Since all purities are at most one, Eq. (10) gives $P_s(n,2\beta) \ge (1-\varepsilon)^3$. Squaring the transfer operator therefore gives $P_s(2n,2\beta) \ge 1-\delta$, where

$$\delta = f\big(1-(1-\varepsilon)^3\big) = \tfrac{1}{2}\big[(1-\varepsilon)^{-3} - 1\big]^2 \le 5\varepsilon^2 \le \varepsilon. \quad (12)$$

The bound uses $1-(1-\varepsilon)^3 \le 3\varepsilon$ and $9/[2(1-3\varepsilon)^2] \le 5$.

This proves the transfer-purity bound in Eq. (8). For the Gibbs-purity bound, insert it into Eq. (11). Using $\delta \le \varepsilon$ gives $P_\tau(4n,\beta) \ge (1-\varepsilon)^3$. Squaring the Gibbs operator then gives $P_\tau(4n,2\beta) \ge 1-\delta$. Thus both bounds in Eq. (8) hold.

From finite input to all lengths.— We now apply the doubling estimate to Proposition 5.5 of Ref. [1]:

$$P_s(2304,784),\ P_\tau(4608,784) \ge 0.99. \quad (13)$$

The energy shift used there cancels from the purity ratios. For $j \ge 0$, set

$$n_j = 2304\,2^j, \qquad \beta_j = 784\,2^j, \qquad \varepsilon_j = \frac{20^{-2^j}}{5}. \quad (14)$$

Since $\varepsilon_0 = 1/100$ and $\varepsilon_{j+1} = 5\varepsilon_j^2$, induction in Eq. (8) gives $P_s(n_j,\beta_j), P_\tau(2n_j,\beta_j) \ge 1-\varepsilon_j$. The error bound is exponential in inverse temperature: $\varepsilon_j = \frac{1}{5}e^{-\beta_j(\log 20)/784}$.

To include every even $L \ge 2304$, choose j with $n_j \le L \le 2n_j$. The moment formula (6) and monotonicity of ℓ^p norms give

$$Z_L(\beta_j) \le Z_{n_j}(\beta_j)^{L/n_j},$$
$$Z_L(2\beta_j) \ge Z_{2n_j}(2\beta_j)^{L/(2n_j)}. \quad (15)$$

Substituting into the Gibbs-purity ratio (3) yields

$$P_\tau(L,\beta_j) \ge \big[P_\tau(2n_j,\beta_j)P_s(n_j,\beta_j)^2\big]^{L/(2n_j)}$$
$$\ge (1-\varepsilon_j)^3 \ge 1 - 3\varepsilon_j > \tfrac{1}{2}. \quad (16)$$

Here both purities in brackets are at least $1-\varepsilon_j$, and $L/(2n_j) \le 1$. This is the bound needed in Eq. (4):

$$e^{-\beta_j \Delta_L} \le \frac{3\varepsilon_j}{1-3\varepsilon_j} \le \frac{60}{97}\,20^{-2^j} < 20^{-2^j}. \quad (17)$$

Taking logarithms and dividing by $\beta_j = 784\,2^j$ proves

$$\boxed{\Delta_L > \frac{\log 20}{784} \simeq 0.003821 \quad (L \ge 2304 \text{ even}).} \quad (18)$$

Consequently, the ground state is unique for these lengths and $\liminf_{L\to\infty,\ L\,\text{even}} \Delta_L > 0$.

We thank Shengqi Sang, Yimu Bao, and Xiao-Liang Qi for helpful discussions.

AI-use declaration.— S. L. prompted GPT-6-Astra (OpenAI) to recast the proof of Ref. [1] in tensor-network diagrams.

[1] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Proposition 5.5 and Appendices A–C.

[2] W. Tang, H.-H. Tu, and L. Wang, Continuous matrix product operator approach to finite temperature quantum states, Phys. Rev. Lett. **125**, 170604 (2020).

[3] B. Pirvu, V. Murg, J. I. Cirac, and F. Verstraete, Matrix product operator representations, New J. Phys. **12**, 025012 (2010).